

DATABASE DESIGN DESCRIPTION

1. DATABASE MANAGEMENT SYSTEM (DBMS) SELECTION

Selected DBMS: PostgreSQL

Reason for selection:

In the Physical Data Modelling phase³, our team selected PostgreSQL over SQLite or MySQL for the AIMS system for the following reasons:

1. **Advanced Data Types:** The design requires storing complex structures such as track lists for CDs. PostgreSQL's native support for JSONB allows efficient storage and querying of semi-structured data.
2. **Inheritance Support:** The system utilizes a **Class Table Inheritance** strategy for the Product domain (Book, CD, DVD, Newspaper). PostgreSQL handles Foreign Key constraints robustly, ensuring referential integrity between parent and child tables.
3. **Concurrency & Scalability:** Unlike SQLite, which is file-based and limited in concurrent write operations⁴, PostgreSQL is a powerful client-server RDBMS capable of handling simultaneous transactions (Orders, Payments) required by a real-world e-commerce environment.

2. PHYSICAL DATA MODEL

Below is the detailed description of the tables in the physical database, formatted according to the project documentation standards⁵.

2.1. DOMAIN: USER & CART

Table: app_user

Stores registered user information.

#	PK	FK	Column Name	Data Type	Mandatory	Description
1	X		user_id	VARCHAR	Yes	Unique user identifier
2			name	TEXT	No	Full name of the user
3			email	TEXT	Yes	User email (Unique)
4			phone_number	TEXT	No	Contact number
5			role	TEXT	No	Permission role (e.g., ADMIN)
6			is_blocked	BOOLEAN	No	Account status flag
7			created_at	TIMESTAMPZ	No	Registration timestamp

Table: cart

Represents the user's shopping session.

#	PK	FK	Column Name	Data Type	Mandatory	Description
1	X		cart_id	VARCHAR	Yes	Unique cart identifier
2		X	user_id	VARCHAR	Yes	Reference to app_user
3			created_at	TIMESTAMPTZ	No	Session start time
4			last_updated_at	TIMESTAMPTZ	No	Last activity time
5			total_amount	BIGINT	No	Cached total price
6			total_weight	DOUBLE PRECISION	No	Cached total weight

Table: cart_item

Individual products inside a cart.

#	PK	FK	Column	Data	Mandatory	Description
---	----	----	--------	------	-----------	-------------

			Name	Type	ry	on
1	X		cart_item_id	VARCHAR	Yes	Unique line item ID
2		X	cart_id	VARCHAR	Yes	Referenc e to cart
3		X	product_id	VARCHAR	Yes	Referenc e to product
4			quantity	INTEGER	No	Number of items
5			unit_price	BIGINT	No	Price at time of adding
6			sub_total	BIGINT	No	Calculate d total

2.2. DOMAIN: ORDER & PAYMENT

Table: customer_order

Stores order headers.

#	PK	FK	Column Name	Data Type	Mandato ry	Descripti on
1	X		order_id	VARCHAR	Yes	Unique

						order ID
2		X	user_id	VARCHAR	Yes	Reference to app_user
3			created_at	TIMESTAMPZ	No	Order placement time
4			status	VARCHAR	No	Order lifecycle status
5			payment_status	VARCHAR	No	Payment confirmation status
6			total_product_price	BIGINT	No	Sum of item prices
7			vat_amount	BIGINT	No	Tax amount
8			shipping_fee	BIGINT	No	Delivery cost
9			total_amount_to_pay	BIGINT	No	Final billable amount

Table: order_item

Snapshot of items in a finalized order.

#	PK	FK	Column Name	Data Type	Mandatory	Description
1	X		order_item_id	VARCHAR	Yes	Unique ID
2		X	order_id	VARCHAR	Yes	Reference to customer_order
3		X	product_id	VARCHAR	Yes	Reference to product
4			quantity	INTEGER	No	Purchased quantity
5			unit_price	BIGINT	No	Locked price per unit
6			sub_total	BIGINT	No	Line total

Table: delivery_info

Shipping details for a specific order.

#	PK	FK	Column Name	Data Type	Mandatory	Description
1	X		delivery_info_id	VARCHAR	Yes	Unique ID

2		X	order_id	VARCHAR	Yes	Reference to customer_order (Unique)
3			receiver_name	TEXT	No	Recipient Name
4			phone_number	TEXT	No	Recipient Phone
5			address_line	TEXT	No	Street Address
6			city	TEXT	No	City
7			province	TEXT	No	Province/State
8			region_type	VARCHAR	No	Used for shipping calc
9			shipping_fee	BIGINT	No	Specific fee for this delivery

Table: payment_transaction

Records of payment attempts.

#	PK	FK	Column Name	Data Type	Mandatory	Description
---	----	----	-------------	-----------	-----------	-------------

1	X		transacti on_id	VARCHAR	Yes	Unique ID
2		X	order_id	VARCHAR	Yes	Referenc e to customer _order
3		X	user_id	VARCHAR	Yes	Referenc e to app_user
4			amount	BIGINT	No	Transacti on value
5			method	VARCHAR	No	Payment method (e.g., QR, Card)
6			status	VARCHAR	No	Success/ Fail status
7			created_ at	TIMESTA MPTZ	No	Transacti on timestam p
8			message	TEXT	No	Respons e message
9		X	vietqr_id	INTEGER	No	Referenc e to vietqr config

Table: vietqr

Configuration for QR code payments.

#	PK	FK	Column Name	Data Type	Mandatory	Description
1	X		vietqr_id	SERIAL	Yes	Auto-increment ID
2			provider_name	TEXT	No	Bank Name
3			bank_code	TEXT	No	Bank Code (Bin)
4			account_number	TEXT	No	Merchant Account No
5			merchant_id	TEXT	No	Merchant Identifier
6			callback_url	TEXT	No	Webhook URL

Table: credit_card_info

Details for card transactions.

#	PK	FK	Column Name	Data Type	Mandatory	Description
1	X		credit_ca	SERIAL	Yes	Auto-incr

			rd_info_id			ement ID
2		X	transacti on_id	VARCHAR	Yes	Referenc e to payment_ transacti on
3			card_nu mber	TEXT	No	Masked Card Number
4			card_hol der_nam e	TEXT	No	Name on Card
5			expiry_m onth	SMALLIN T	No	Expiratio n Month
6			expiry_ye ar	SMALLIN T	No	Expiratio n Year
7			cvv	TEXT	No	Security Code

2.3. DOMAIN: PRODUCT (Class Table Inheritance)

Table: product (Base Entity)

Contains attributes shared by all product types.

#	PK	FK	Column Name	Data Type	Mandato ry	Descripti on
---	----	----	----------------	--------------	---------------	-----------------

1	X		product_id	VARCHAR	Yes	Universal Product ID
2			title	TEXT	No	Product Name
3			category	TEXT	No	Category
4			original_value	BIGINT	No	Import Price
5			current_price	BIGINT	No	Selling Price
6			weight	DOUBLE PRECISION	No	Weight (kg)
7			height	DOUBLE PRECISION	No	Height (cm)
8			length	DOUBLE PRECISION	No	Length (cm)
9			width	DOUBLE PRECISION	No	Width (cm)
10			barcode	TEXT	No	Barcode String
11			status	INTEGER	No	Availability Status
12			description	TEXT	No	Product Details

13			img_url	TEXT	No	Image Path
----	--	--	---------	------	----	------------

Table: book

Extends product.

#	PK	FK	Column Name	Data Type	Mandatory	Description
1	X	X	product_id	VARCHAR	Yes	FK referencing product
2			cover_type	TEXT	No	Hardcover/Paperback
3			nb_pages	INTEGER	No	Number of pages
4			genre	TEXT	No	Literary Genre
5			publisher	TEXT	No	Publisher Name
6			publication_date	DATE	No	Date Published
7			language	TEXT	No	Language

8			authors	TEXT	No	Author Name(s)
---	--	--	---------	------	----	----------------

Table: cd

Extends product.

#	PK	FK	Column Name	Data Type	Mandatory	Description
1	X	X	product_id	VARCHAR	Yes	FK referencing product
2			artist	TEXT	No	Artist Name
3			record_label	TEXT	No	Record Label
4			music_type	TEXT	No	Genre
5			release_date	DATE	No	Release Date
6			tracks	JSONB	No	List of Tracks (JSON)

Table: dvd

Extends product.

#	PK	FK	Column Name	Data Type	Mandatory	Description
1	X	X	product_id	VARCHAR	Yes	FK referencing product
2			disc_type	TEXT	No	Format (e.g., Blu-ray)
3			director	TEXT	No	Director Name
4			runtime	INTEGER	No	Duration (minutes)
5			studio	TEXT	No	Studio Name
6			subtitle	TEXT	No	Subtitles
7			release_date	DATE	No	Release Date

Table: newspaper

Extends product.

#	PK	FK	Column Name	Data Type	Mandatory	Description
1	X	X	product_id	VARCHAR	Yes	FK

			d			referenci ng product
2			publisher	TEXT	No	Publisher
3			editor_in_ chief	TEXT	No	Editor Name
4			issue_nu mber	TEXT	No	Issue Number
5			publicati on_date	DATE	No	Date Publishe d
6			publicati on_frequ ency	DOUBLE PRECISIO N	No	Frequenc y
7			language	TEXT	No	Languag e
8			issn	TEXT	No	ISSN Code
9			sections	TEXT	No	Newspap er Sections